

Project Overview: Linear Regression and Model Diagnostics

1. Significance of Predictors in Multiple Linear Regression

Using the Advertising dataset (Table 3.4), a multiple linear regression model was used to predict sales based on advertising budgets across three channels: TV, radio, and newspaper.

Key Findings:

- TV and radio had very low p-values ($p < 0.0001$), indicating strong statistical evidence that both are significantly related to sales.
- Newspaper had a high p-value (≈ 0.86), suggesting no significant relationship between newspaper advertising and sales.
- Conclusion: Firms should focus advertising spend on TV and radio, as newspaper advertising is unlikely to impact sales meaningfully.

2. Interaction Effects and Model Interpretation

(a) Correct Interpretation:

An interaction between GPA and education level affects salary. When GPA is sufficiently high, high school graduates may earn more than college graduates due to the negative interaction effect.

(b) Salary Prediction:

A college graduate with GPA = 4.0 and IQ = 110 is predicted to earn \$137.10 based on the given linear model.

(c) Misconception Clarified:

Even a small interaction coefficient (e.g., 0.01) can be statistically significant. The magnitude alone does not determine importance—p-values and context matter.

3. Simple Linear Regression Properties

This section proved that the least squares regression line always passes through the point (mean of x , mean of y). A numerical example in R confirmed that the predicted y at the mean of x equals the mean of y .

4. Modeling with the Auto Dataset

Using the Auto dataset, a simple linear regression model was fitted with horsepower as the predictor of mpg.

Results:

- Negative coefficient (-0.158): Suggests mpg decreases with increasing horsepower.
- $R^2 = 0.61$: Horsepower explains about 61% of the variation in mpg.
- Predictions: At horsepower = 98, mpg is estimated to be 24.47, with confidence and prediction intervals included.

Model Diagnostics:

- Residual plots indicated a reasonable fit.
- QQ plot suggested approximate normality of residuals.
- Slight heteroscedasticity observed in Scale-Location plot.
- Some high-leverage points identified, warranting further inspection.

5. No-Intercept Regression and Symmetry

Key Insight:

- Regressing y on x (no intercept) and x on y both yielded high t-values and similar significance.
- This symmetry confirms the strength of the linear relationship regardless of regression direction.

Manual t-statistic Verification:

- A manually computed t-statistic matched the model output, confirming reliability.

6. Multicollinearity and Simulation Study

A simulated dataset was created with multicollinearity between predictors x1 and x2.

Key Results:

- x1 and x2 were highly correlated ($r \approx 0.835$).
- In the full model ($y \sim x1 + x2$):
 - Standard errors were inflated.
 - x2 became non-significant despite its true effect.

- In individual models:
 - x_1 and x_2 appeared more significant but produced biased estimates due to omitted variable bias.

7. Influence of High-Leverage Points

A new data point was added, and the models were re-evaluated:

- The outlier had a strong impact on coefficients and standard errors.
- Diagnostic plots confirmed it as a high-leverage influential point, highlighting model sensitivity.

8. Statistical Consistency Through Simulation

The final section emphasized that if the data generation process is repeated many times:

- The average estimated coefficients converge to the true values.
- This consistency reflects the unbiased nature of least squares estimation and the law of large numbers.

Conclusion

This assignment thoroughly explored linear regression from both theoretical and practical perspectives:

- Assessed variable importance via p-values and R^2 .
- Investigated interaction effects and model interpretation.
- Diagnosed model assumptions using visual residual checks.
- Examined multicollinearity and leverage, and their impact on model stability.
- Reinforced theoretical principles with simulation and analytical proofs.

These insights equip analysts with a stronger foundation to apply regression responsibly, interpret results rigorously, and detect potential model pitfalls in practice.